

Practical No.4-A Write a CUDA Program for the addition of two large vectors

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Vectoradd.cu -

```
#include <stdio.h>

#include <cuda_runtime.h>

#define N 1000000 // 1 Million elements (large vector)

__global__ void vectorAdd(float *a, float *b, float *c, int n) {

    int tid = blockIdx.x * blockDim.x + threadIdx.x;

    if (tid < n) {

        c[tid] = a[tid] + b[tid];

    }

}

int main() {

    float *h_a, *h_b, *h_c; // Host arrays

    float *d_a, *d_b, *d_c; // Device arrays

    size_t size = N * sizeof(float);

    // Allocate host memory

    h_a = (float*)malloc(size);

    h_b = (float*)malloc(size);

    h_c = (float*)malloc(size);

    // Initialize vectors

    for (int i = 0; i < N; i++) {

        h_a[i] = (float)i;

        h_b[i] = (float)i * 2.0f;

    }

    // Allocate device memory

    cudaMalloc(&d_a, size);

    cudaMalloc(&d_b, size);

    cudaMalloc(&d_c, size);
```

```

// Copy data Host → Device
cudaMemcpy(d_a, h_a, size, cudaMemcpyHostToDevice);
cudaMemcpy(d_b, h_b, size, cudaMemcpyHostToDevice);

// Launch kernel
int threadsPerBlock = 256;
int blocksPerGrid = (N + threadsPerBlock - 1) / threadsPerBlock;
vectorAdd<<<blocksPerGrid, threadsPerBlock>>>(d_a, d_b, d_c, N);

// Copy result Device → Host
cudaMemcpy(h_c, d_c, size, cudaMemcpyDeviceToHost);

// Print first 10 results
printf("Vector Addition Results (First 10 elements):\n");
for (int i = 0; i < 10; i++) {
    printf("%.0f + %.0f = %.0f\n", h_a[i], h_b[i], h_c[i]);
}

// Optional: Check last element
printf("...\nLast element: %.0f + %.0f = %.0f\n",
        h_a[N-1], h_b[N-1], h_c[N-1]);

// Free memory
free(h_a); free(h_b); free(h_c);
cudaFree(d_a); cudaFree(d_b); cudaFree(d_c);

printf("\nVector addition completed successfully on GPU!\n");
return 0;
}

```

Output:

vedant@LAPTOP-2AJF2JRG: nvcc -o vectoradd vectoradd.cu

vectoradd.cu

tmpxft_000062ac_00000000-7_vectoradd.cudafe1.cpp

vedant@LAPTOP-2AJF2JRG: vectoradd.exe

Vector Addition Results (First 10 elements):

0 + 0 = 0

$$1 + 2 = 3$$

$$2 + 4 = 6$$

$$3 + 6 = 9$$

$$4 + 8 = 12$$

$$5 + 10 = 15$$

$$6 + 12 = 18$$

$$7 + 14 = 21$$

$$8 + 16 = 24$$

$$9 + 18 = 27$$

...

$$\text{Last element: } 999999 + 1999998 = 2999997$$

Vector addition completed successfully on GPU!